

VICTOR IBHAFIDON

Simulation & Digital Twin Consultant

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PROFESSIONAL SUMMARY

Engineer with technical depth across security, data and cloud, and a proven ability to consult with stakeholders to map products and systems to business use cases. At Bastion Shield Technologies I work with clients and security teams on AI governance, turning requirements into solutions. I own work end to end, understanding what can go wrong and proving it works, and communicate clearly with both technical and non-technical audiences. I have worked closely with statistical and machine-learning models in production, integrating trained models into live runtimes for real-time scoring and evaluating them against baselines rather than building them in isolation.

TECHNICAL SKILLS

Backend & Data: Data Pipelines, FastAPI, Django, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, pgvector, Vector Search, ETL

Security & AI Governance: AI Security, AI Governance, Authentication, Authorization, RBAC, API Security, OWASP Top 10, Prompt Injection Mitigation, AI Safety, Responsible AI, GDPR

AI & Machine Learning: LLMs, Generative AI, AI Agents, Multi-Agent Systems, LangGraph, MCP, JSON-RPC, Tool Calling, RAG, GraphRAG, Prompt Engineering, AI Evaluation, PyTorch, Transformers, Scikit-learn, Computer Vision, NLP

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, EKS), Microsoft Azure, Azure OpenAI, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Nginx, OpenTelemetry, Prometheus, Grafana

Programming: Python, TypeScript, JavaScript, SQL, Bash

MLOps & Model Operations: MLflow, Model Training, Model Evaluation, Model Monitoring, Drift Detection, Experiment Tracking, Model Deployment

Consulting & Advisory: Solutions Consulting, Stakeholder Consultation, Use-Case Mapping, Requirements Gathering, Presentations, Client Communication, Discovery Workshops, Recommendations

Domain Knowledge: AI Governance, Cyber Security, Data Platforms, Cloud (AWS), Compliance (GDPR)

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML), Bastion Shield Technologies

London, United Kingdom | April 2026 – Present

- My work at Bastion Shield Technologies is fundamentally consultative: understanding what clients are trying to achieve, mapping Bulwark, an AI governance platform covering agent identity, policy enforcement, detection of unregistered agents and compliance, to their use case, and guiding them through design and delivery. I turn business requirements into working, measurable solutions, and explain complex technical trade-offs clearly to non-technical stakeholders.
- Engineered the backend in Python, TypeScript, and PostgreSQL, implementing RBAC, server-side tenant isolation, secure authentication, API-key management, and an SDK for external teams to register agents and stream policy decisions. Designed tamper-evident audit logging with offline verification, strengthened data privacy controls, and remediated four production vulnerabilities, including a session-token forgery path.
- Developed behavioural anomaly detection combining statistical methods with usage forecasting to identify unusual agent activity. The system achieved a 0.981 ROC-AUC against an Isolation Forest baseline and was integrated into the TypeScript runtime for real-time scoring.

Full Stack Software Engineer, Basketball Nxtion

London, United Kingdom | February 2026 – April 2026

- Developed the MVP of Pplyground, a mobile application connecting basketball players, creators, and sports communities, contributing across product design, application architecture, backend development, APIs, authentication, and user workflows.
- Designed and integrated an AI architecture connecting LLM/ML services with PostgreSQL data, backend APIs, and application business logic. Built workflows for personalised recommendations, intelligent user matching, and content discovery, using user and platform data to generate relevant results, improve engagement, and automate discovery workflows.
- Designed the PostgreSQL data architecture and analytics pipelines for the MVP, structuring user, activity, engagement, and platform data to support data ingestion, transformation, analysis, and business intelligence. Built data workflows enabling insights into platform performance, user behaviour, engagement, and product adoption.

Full Stack Python Engineer Intern, CyberAI Technologies

London, United Kingdom | September 2025 – November 2025

- Collaborated with senior AI engineers to develop Surfyn AI, a conversational AI customer support platform enabling businesses to deploy knowledge-based AI assistants grounded in websites and documents to automate customer interactions and provide relevant, context-aware responses.
- Contributed to the development of RAG and conversational AI workflows using Python and Django, working on document ingestion, chunking, embeddings, vector-based retrieval, prompt construction, conversation context, REST APIs, and LLM integrations while learning how knowledge-based AI systems are designed and evaluated.
- Developed backend functionality and AI workflow improvements, working on API development, debugging, testing, performance optimisation, knowledge retrieval, and integration between AI services and the Django application.

SELECTED PROJECTS

Portivo Estate Suite (AI Powered PropTech Operating System): Approached Portivo from a consulting and product-engineering perspective, working with estate managers to understand how AI could be integrated into existing property operations to reduce administrative workload and improve lettings, sales, maintenance, compliance and reporting. Worked directly with users through discovery, workflow design, testing and implementation, translating feedback into product requirements, documenting APIs and operational processes, validating end-to-end workflows and supporting customer configuration and adoption. Designed the system architecture and took ownership of end-to-end development across product analysis, UX/UI, frontend, backend, AI/ML, data engineering, cloud, security, QA, AI governance and integrations. Built the multi-tenant application with React, TypeScript, Next.js and Tailwind, developing dashboards, property management, CRM, reporting and administrative workflows, backed by Python, FastAPI and SQLAlchemy for business logic and APIs. Designed the PostgreSQL data model with tenant-scoped queries and Alembic migrations against a live customer database, using Redis and Celery for caching, asynchronous processing and scheduled jobs. Built the AI architecture using LangGraph, Anthropic Models, RAG, OCR, chunking, embeddings and pgvector, with an orchestrator delegating typed tasks to specialist agents for document retrieval, lease abstraction, arrears detection, compliance tracking, maintenance triage and outbound sales prospecting. Implemented MCP servers and scoped credentials to give agents controlled access to CRM, email and document tools rather than creating bespoke integrations. Evaluated AI performance using labelled datasets for groundedness and extraction accuracy, with Langfuse for tracing, schema-validated outputs, confidence thresholds and audit trails for agent actions. Applied RBAC, tenant isolation, scoped permissions, security controls and AI governance to protect customer data and control agent behaviour, including prompt-injection and data-leakage risks. Containerised and deployed the platform to the cloud using Docker and GitHub Actions CI/CD, while designing API integrations, technical documentation and implementation workflows to support enterprise adoption. **Live in production at:** portivoestatesuite.com

Autonomous AI Defence System for Critical Infrastructure: Designed an autonomous surveillance and anomaly-detection system for critical infrastructure such as rail corridors, electrical substations, solar and wind farms, ports, reservoirs and industrial sites. Drones and ground vehicles can run scheduled or operator-triggered missions in simulation, using computer vision to detect intrusions, trespassing, vegetation encroachment and equipment anomalies. Designed the architecture as an event-driven system where simulated vehicles generate telemetry and camera data, computer vision services process observations, and Kafka distributes events to the backend for alerting, tracking and decision-making. Built FastAPI services and PostgreSQL for mission, asset and operational state, Redis for fast-access state and coordination, and Next.js/WebSockets for a live operations dashboard showing vehicle locations, telemetry, alerts and mission status. Designed a bounded autonomy layer that evaluates detected events against predefined rules and operational constraints before proposing actions, with high-impact actions routed to a human operator for approval, override or emergency stop. Integrated YOLO/PyTorch for computer vision inference and designed a hardware integration layer using MQTT and ROS to connect mission and control services with drones and ground vehicles through standardised telemetry, command and sensor interfaces, separating vehicle hardware from the core application and allowing the same architecture to support simulation and physical robotics hardware. Added an audit layer that records missions, detections, decisions, approvals and commands so actions can be reconstructed and reviewed. Designed the ML lifecycle around model training, evaluation, fine-tuning, drift detection and retraining triggers, with Prometheus/Grafana for system and model observability. Containerised the services with Docker and designed Kubernetes, Terraform and AWS infrastructure for repeatable deployment, while implementing JWT, RBAC, OWASP-aligned protections, input validation, SSRF/injection safeguards and explicit failure-mode handling. Model integration was a core part of this work: trained YOLO and PyTorch models were integrated into the running system for inference, with MLOps covering model evaluation, drift detection and retraining so model behaviour stayed measurable after deployment. GitHub: github.com/Web8080/Autonomous_AI_defense_system_design

EDUCATION

MSc Data Science, Nottingham Trent University, United Kingdom (2023 – 2024)

BSc Computer Science, Ambrose Alli University, Nigeria (2016 – 2021)